Defect-aware graph learning of impurity incorporation energetics and chemical transferability in two-dimensional materials

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Machine-learned formation energies can accelerate impurity screening in two-dimensional materials, but random-split accuracy alone does not establish chemical transferability or a reliable deployment domain. We develop a defect-aware graph transformer and evaluate it on a provenance-audited, leakage-controlled subset of the IMP2D database. A paired 2^3 factorial separates the contributions of gated graph readout, defect-environment features, and a pre-normalized, distance-gated local interaction block. The architecture is selected using validation data only and is then tested under random, host-held-out, impurity-held-out, host–impurity-pair-held-out, and predefined compositional-block protocols against periodic SchNet and tuned descriptor baselines. A separate calibration partition supports ensemble uncertainty and split-conformal intervals, while out-of-fold predictions quantify defect-type preference and site-screening regret. The resulting analysis is designed to distinguish interpolation accuracy from chemically meaningful transfer and to state the applicability limits supported by IMP2D. External first-principles validation is outside the present scope.

I. INTRODUCTION

Point defects and incorporated impurity atoms can alter carrier density, localized electronic states, magnetism, and catalytic activity in two-dimensional (2D) materials. Formation energy is therefore a central quantity in computational defect screening: for a specified host, impurity reservoir, charge state, and structural configuration, it ranks the energetic cost of incorporation. Exhaustive first-principles evaluation remains expensive because the search space grows across host chemistry, impurity identity, incorporation class, and symmetry-distinct site.

The IMP2D database provides a systematic first-principles survey of neutral adsorbate and interstitial impurities in 2D hosts [1]. Its scale makes supervised surrogates possible, and graph neural networks are natural models because they operate directly on periodic atomic structures. ALIGNN demonstrated the value of joint bond and angle representations for crystalline properties [2], while SchNet established continuous-filter message passing from interatomic distances [3]. Defect-specific learning has subsequently developed along several complementary directions. Bulk defect energetics have been treated with defect graph networks, semiconductor graph models, persistent-homology features, and charge-aware predictors [4–7]. For 2D systems, sparse defect representations have been evaluated on the 2DMD data, whereas tree and descriptor models have been applied directly to IMP2D [8–10]. The two IMP2D studies show in particular that data representation and model selection materially affect the reported error. These developments motivate architectures that expose the impurity site while retaining both local coordination and longer-range periodic context.

Benchmark accuracy, however, does not by itself define a useful materials model. A random structure split can place closely related host–impurity chemistries, or even numerically equivalent structures, on both sides of the train–test boundary, a known source of optimistic materials-learning estimates [11]. It also does not answer whether a model transfers to an unseen host or impurity, whether a reported architectural gain survives paired repeats, or whether prediction uncertainty identifies a subset on which screening risk is acceptably low. Structure-based out-of-distribution studies have emphasized that materials models can degrade sharply when test chemistry differs from training chemistry [12, 13]. For defect screening, host and impurity axes must therefore be separated explicitly. Uncertainty studies in atomistic machine learning have likewise shown that ensemble disagreement can provide a useful credibility signal, while calibration and out-of-domain reliability remain distinct requirements [14–16]. A defect-screening surrogate should therefore be judged by its transfer partitions, uncertainty calibration, and decision-level screening consequences rather than by one random-split error.

This work asks three connected questions. First, do defect-centered readout, explicit local-environment features, and a distance-gated local interaction block provide reproducible improvements when tested as a complete paired factorial rather than as sequential additions? Second, how does the selected model compare with periodic SchNet and validation-tuned descriptor baselines when hosts, impurities, host–impurity pairs, or a compositional block are held out? Third, can a separately calibrated ensemble support selective prediction and recover the preferred incorporation class or lowest-energy site with low screening regret?

To answer these questions, we construct a fully auditable protocol around the IMP2D source database. We replay the source filter and formation-energy formula, remove rows whose raw energy components cannot be reconstructed, exclude same-element structures for which the relaxed database cannot assign a permutation-invariant impurity node, merge duplicate structures with both coordinate and invariant-distance fingerprints, and publish immutable split definitions. The resulting 10,224 structure modeling set is evaluated through a paired 2^3 architecture experiment, random and chemistry-grouped cross-validation, and a predefined host–impurity matrix holdout. Architecture and descriptor choices use validation data only. A distinct calibration partition is used for variance scaling and split-conformal intervals, and all materials-facing conclusions are computed from out-of-fold predictions.

The intended contribution is consequently broader than a new point estimate on a familiar random split. It is a test of which defect-aware modules are supported by paired evidence, where chemical transfer fails, and how that failure should constrain use of the surrogate. No external first-principles test set is used; the scope is the accuracy, transferability, calibration, and screening utility that can be established from the audited IMP2D evidence.

II. METHODS

A. Dataset, target provenance, and canonical modeling set

We use neutral impurity structures from the Impurities in Two-Dimensional Materials Database (IMP2D) [1]. The source ASE database contains 17,364 rows. Replaying the data-preparation filter in database order removes 4551 unconverged calculations, 1828 rows with a missing or nonfinite stored formation energy, and 344 rows with $|E_\mathrm{f}| > 20$ eV, leaving 10,641 structures. These structures span 44 hosts, 65 impurity elements, and two incorporation classes (adsorbate and interstitial).

For rows with complete source fields, the target was independently recomputed as

$$E_\mathrm{f} = E_\mathrm{def} - E_\mathrm{host} - \mu_\mathrm{imp}. \tag{1}$$

The recomputed and stored values agree to a maximum absolute numerical difference of 2.3×10^{-13} eV. Four otherwise filtered rows contain a zero sentinel in the stored defective-structure energy and therefore cannot be independently reconstructed from the source components; these rows are excluded from every modeling split.

The released relaxed structures do not retain a persistent identity field for the atom initially inserted by the structure generator. In 349 rows, the impurity element is also a constituent of the host, so multiple identical nuclei match the impurity label. Assigning one of them by array order would break permutation invariance and would make the defect-centered features depend on a nonphysical convention. We therefore require exactly one atom to match the impurity element and exclude all 349 nonunique rows from the formal protocol.

We also prevent symmetry-equivalent structures from crossing split boundaries. Duplicate candidates are identified by the union of two hashes: (i) atomic numbers, cell, and Cartesian coordinates rounded to 10^{-6}, and (ii) a permutation- and translation-invariant spectrum of element-labeled periodic pair distances rounded to 10^{-4} Å. The latter detects numerically equivalent structures even when atom order or coordinate origin differs. Among the remaining eligible structures, connected duplicate groups contain 64 redundant rows; the lowest-index auditable representative is retained. The resulting canonical modeling set contains 10,224 structures. The complete audit, excluded sample indices, database hashes, and immutable split files are distributed with the code.

B. Fixed evaluation partitions

All partitions were generated once from the canonical set and stored as index lists. Five paired random 80/10/10 partitions (assignment seeds 42–46) are used for the 2^3 architecture experiment. A separate fivefold random cross-validation partition provides one out-of-fold prediction for every retained sample. Chemical transfer is evaluated using balanced fivefold group cross-validation in which complete host identities, impurity identities, or host–impurity pairs are held together. In each grouped fold, one group fold is used for testing and the next group fold for validation. We further use a predefined matrix-completion holdout: group-VI dichalcogenide hosts combined with $3d$ impurities form the test block, while group-IV/V hosts combined with $4d$ impurities form the validation block. After the common audit exclusions, this partition contains 9,487 training, 372 validation, and 365 test structures.

Uncertainty quantification uses an independent random partition with 75% for training, 10% for checkpoint selection, 5% for calibration, and 10% for final testing (assignment seed 62). The calibration and test indices are not used to fit network parameters or select checkpoints. All split files cover the full 10,641-row container through mutually disjoint train, validation, test, optional calibration, and exclusion sets; their SHA256 hashes are stored in every run manifest.

Figure 1 summarizes the sequential data audit, canonical chemical coverage, target distribution, and fixed partition geometry.

C. Defect-aware graph transformer

Each periodic supercell is represented by an atomistic graph. We refer to the resulting model as the Defect-Aware Radial Transformer (DART). The initial node representation concatenates a fixed 128-dimensional ct-UAE elemental vector with nine elemental attributes: group, period, Pauling electronegativity, covalent and van der Waals radii, a group-based valence count, first ionization energy, electron affinity, and atomic mass [17]. The nine columns are standardized over atomic numbers 1–100. An unavailable van der Waals radius is replaced by

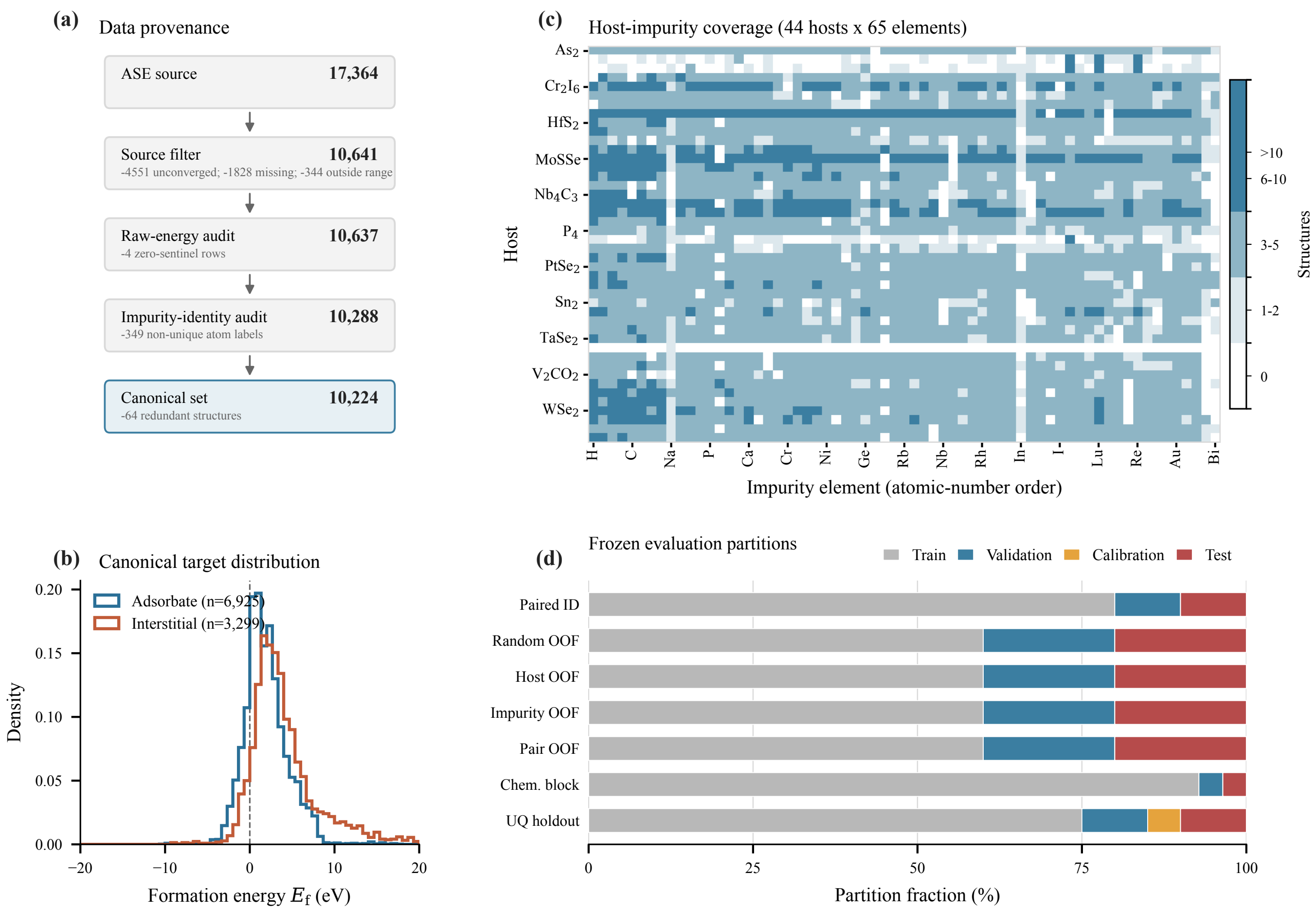

FIG. 1. Data provenance and frozen evaluation protocol. (a) Sequential source filtering, raw-energy reconstruction audit, impurity-identity audit, and structure canonicalization. The final set excludes four rows with zero sentinels in a raw energy component, 349 rows without a unique impurity atom, and 64 redundant structures. (b) Formation-energy distributions for the two incorporation classes in the canonical set. (c) Number of structures for each of 44 hosts and 65 impurity elements; white cells are unobserved combinations. (d) Train, validation, calibration, and test fractions for each evaluation regime. Bars for fivefold protocols show fold averages. All regimes cover the same 10,224 retained structures; the common 417-row exclusion set is omitted from the bars.

the corresponding covalent radius; the other tabulated values, including explicit unavailable or approximate entries, are fixed by the released feature table. We reconstruct a 100-element table from the published checkpoint by applying its linear `atom_embed` layer to each one-hot elemental code, $\mathbf{U} = \mathbf{W}^{\mathsf{T}} + \mathbf{1}\mathbf{b}^{\mathsf{T}}$. A zero padding row is prepended internally, so atomic numbers 1–100 address rows 0–99 of $\mathbf{U}$ without an index shift. The concatenated representation is projected to 128 hidden channels, and a learned binary embedding marks the impurity atom. Three local interaction layers aggregate neighbors within 5 Å. Distances are expanded in 32 Gaussian radial basis functions, and three-body messages use a 32-function angular expansion. Two subsequent four-head geometric Transformer blocks apply self-attention with a learned radial bias derived from the periodic pair-distance matrix on a radial grid ending at 12 Å [18]. A two-layer multilayer perceptron maps the pooled graph representation to $\hat{E}_f$. Because every formal row passes the identity audit, the binary impurity embedding always marks exactly one compositionally unique atom; no array-order heuristic is used.

The confirmatory architecture experiment varies three binary modules while holding every other training and model setting fixed:

Gated readout (G).: A learned atom-attention sum and a channelwise maximum are fused by a graph-dependent scalar gate. The off state uses masked mean pooling.

Environment enrichment (E).: Four defect-centered quantities—coordination number, mean and maximum neighbor distance, and the absolute electronegativity contrast to the local neighborhood—are normalized, projected, and added only to the impurity representation. The off state omits this branch.

Pre-norm gated local block (P).: The on state normalizes node states before message construction and applies a learned scalar distance gate to each edge before a residual update. The off state uses the original ungated message and post-residual normalization [19]. We treat this as one composite local-block change and do not attribute its effect to normalization or edge gating separately.

The full 2^3 factorial contains all eight G/E/P combinations on each of the five paired random partitions. Within a partition, all variants share the same initialization seed and training batches. Main effects and interactions are orthogonal factorial contrasts. The architecture promoted to subsequent transfer and uncertainty experiments is the variant with the lowest mean validation MAE; exact ties are resolved by fewer enabled modules and then by binary variant code. Test metrics are not part of the ordering rule.

D. Training protocol

Networks are optimized for 150 epochs with AdamW, a peak learning rate of 5×10^{-4} reached after a ten-epoch linear warmup from 5×10^{-5}, weight decay 10^{-4}, and cosine decay to 10^{-6}. The objective is mean absolute error (MAE), batch size is 64 for DART, and gradient norms are clipped at 5. Targets are standardized using training-partition statistics for optimization and transformed back to eV for all reported errors. The fixed DART regularization recipe is limited to 0.1 dropout and additive Gaussian target noise with standard deviation 0.03 eV; no coordinate or strain augmentation is applied. A host-balanced sampler draws 200 structures per host per epoch; its sequence and the independent target-noise stream are deterministically indexed by model seed and epoch so paired variants see the same batches and noise realizations. Every factorial variant uses the same two fixed input assets. The ct-UAE table described above is a non-trainable buffer. Separately, we pretrained the nine-feature input projection and V1 local interaction stack for 30 epochs. We first drew 20,000 records uniformly without replacement from JARVIS-DFT `dft_3d` with seed 42; conversion and graph-validity filters retained 19,902 pristine three-dimensional crystals [20]. Pretraining used formation energy per atom as the source target, an 80/10/10 random partition, and seed 42. No IMP2D target is used in this stage. An explicit chemistry-overlap audit found that 26 of the 44 IMP2D host labels (25 of 42 distinct reduced host formulas) occur among 69 JARVIS source records, and that all 65 IMP2D impurity elements occur somewhere in the source-task structures. Formula overlap does not establish structural identity, but it means that the grouped partitions withhold IMP2D formation-energy supervision rather than all prior source-task exposure to the same reduced host formula or impurity element.

The resulting checkpoint contributes its nine-feature input-projection block and bias together with all 51 shape-matched V1 local-message tensors. The 128 input-projection columns associated with the fixed ct-UAE vectors retain their seeded initialization; for P-enabled variants, the 12 new edge-gate tensors do likewise. All trainable model parameters are then optimized on the assigned IMP2D training partition. Both input assets are verified against SHA-256 digests, and each run manifest records the exact copied and seeded parameter sets.

The checkpoint with minimum validation MAE is evaluated once on its test partition. Each run records the code commit and dirty state, resolved configuration, command, Python/PyTorch/CUDA environment, data and split hashes, seed, partition indices, epoch history, checkpoint, validation and test predictions, and calibration predictions when the split defines them. SHA-256 digests bind the metrics, checkpoint, split-index, and prediction files to that manifest. Before aggregation, collectors verify partition membership and independently recompute all reported regression metrics from the stored targets and predictions. Interrupted jobs resume from complete optimizer, scheduler, model, and random-number-generator state.

E. Baselines

Two baseline classes use exactly the same canonical partitions. First, an 80-dimensional deterministic descriptor uses atomic number, mass, covalent radius, and Pauling electronegativity. Mean, standard deviation, minimum, maximum, and median over all atoms and host atoms contribute 40 entries; impurity values and impurity-minus-host-mean contrasts contribute eight. Eighteen entries encode atom count, elemental diversity, cell lengths, area, volume, mass density, the first six impurity-neighbor distances, two coordination counts, and local-distance mean and standard deviation. Twelve site indicators and two incorporation-class indicators complete the vector. Missing electronegativities use the fixed value 2.0 and fewer than six neighbors are padded at 20 Å. We fit a training-target mean predictor, ridge regression, random forests, histogram gradient boosting, and LightGBM [21]. These model families also permit direct comparison with earlier IMP2D descriptor studies [9, 10]. Hyperparameters within each family are selected by validation MAE only; the best descriptor family used in each reported regime is likewise chosen by mean validation MAE across folds.

Second, a periodic SchNet comparator [3] uses 128 hidden channels, six interaction blocks, 50 Gaussian functions, and a 5 Å cutoff. Its prespecified graph prediction sums learned atomwise scalar contributions, following the additive energy readout of the original SchNet formulation. It uses batch size 32 but otherwise follows the same target normalization, 150-epoch AdamW schedule, host balancing, gradient clipping, and target-noise protocol as DART; it uses a conventional learned atomic-number embedding and no DART pretraining. Random and host–impurity-pair fivefold evaluation uses one prespecified model seed per fold (242 for DART and 342 for SchNet). Host-held-out, impurity-held-out, and chemistry-block evaluation uses three independently prespecified seeds (242–244 for DART and 342–344 for SchNet). Predictions are averaged within each split before pooled out-of-fold metrics are calculated. Consequently, DART–SchNet contrasts compare the complete prespecified learning recipes, including their different input representations, initialization, and graph readouts; they do not isolate a pure backbone-architecture effect. The selected DART has 820,558 trainable parameters, compared with 455,937 for SchNet. For the same reason, "host held out" and "impurity held out" refer specifically to IMP2D target supervision.

F. Held-out uncertainty calibration and screening analysis

Five independently initialized members of the validation-selected DART architecture are trained on the dedicated uncertainty split, following the deep-ensemble construction and its atomistic use as a model-credibility signal [14, 22]. Ensemble mean and sample standard deviation provide the point prediction and a raw model-disagreement proxy. A fixed-seed random permutation (seed 6201) assigns the dedicated calibration rows to two subsets without using target labels. One subset fits a nonnegative scale and floor for

$$\sigma_i = \sqrt{(as_i)^2 + b^2} \tag{2}$$

by Gaussian negative log likelihood; the other subset estimates finite-sample split-conformal quantiles of $|E_{\mathrm{f},i} - \hat{E}_{\mathrm{f},i}|/\sigma_i$ [15, 23]. Under exchangeability, the resulting intervals have finite-sample marginal coverage at each prespecified level; this guarantee does not imply conditional coverage for every host or impurity group. Test labels do not enter either calibration step, and test metrics are computed only after the scale, floor, and interval quantiles are fixed.

For materials-facing analysis, the five host–impurity-pair folds yield one out-of-fold prediction per retained structure. Within each host–impurity pair we compare the minimum predicted and reference energies of adsorbate and interstitial configurations, identify the predicted lowest-energy site, and report preference accuracy and screening regret. Exact-site metrics require at least two candidate sites, whereas top-two accuracy requires at least three; reference minima within 10^{-8} eV are treated as tied, and tied adsorbate–interstitial pairs are excluded from binary preference accuracy. These are predictive associations within IMP2D, not causal mechanism assignments or external DFT validation.

III. EVALUATION PROTOCOL

A. Metrics and statistical analysis

We report MAE, root-mean-square error (RMSE), signed bias, Spearman rank correlation, and R^2. Host- and impurity-macro MAE give each chemical group equal weight, independent of its number of structures. Cross-validation predictions are concatenated only when test folds are disjoint; model seeds are averaged within a fold before concatenation. Low-energy behavior is assessed by MAE for $E_\mathrm{f} \leq 0$, MAE within the true lowest-energy decile of each pooled test regime, and recall of that decile by the predicted lowest-energy decile. These strata are evaluation summaries and are not used for architecture or hyperparameter selection.

For the architecture factorial, each effect is evaluated as the difference between the average metric at positive and negative levels of the associated orthogonal contrast within each paired repeat. Ninety-five-percent intervals are obtained by nonparametric bootstrap over the five repeat-level contrasts. Because only five paired repeats were prespecified, the intervals are accompanied by repeat-level sign consistency and interpreted as descriptive uncertainty summaries rather than large-sample hypothesis tests. For model comparisons on aligned out-of-fold predictions, we use 10,000 paired bootstrap resamples. The resampling unit matches the held-out unit: individual structures for random cross-validation, complete hosts for host-held-out evaluation, complete impurities for impurity-held-out evaluation, and host–impurity pairs for pair-held-out and compositional-block evaluation. Within each resampled chemical unit all associated structures remain together. Screening intervals use 20,000 host–impurity-pair cluster-bootstrap resamples. Adsorbate and interstitial site decisions from the same pair remain together, while point estimates average all eligible decision sets. An interval excluding zero is treated as evidence for a directional difference; otherwise the result is reported as inconclusive rather than equivalent.

Uncertainty evaluation includes empirical interval coverage and mean width at 50%, 80%, 90%, and 95% nominal coverage, Gaussian negative log likelihood, the continuous ranked probability score, Spearman correlation between uncertainty and absolute error, and risk–coverage curves. The area under the risk–coverage curve (AURC) is accompanied by its excess over oracle ordering [24]. All calibration choices use only the dedicated calibration partition.

No new first-principles calculations are used as external validation in this study. Predictive and screening claims rely exclusively on the canonical IMP2D targets; an independently generated, target-matched external test set is outside the present scope.

IV. RESULTS

A. Validation-locked architecture selection

The complete paired factorial is reported in Table I and Fig. 2. Validation MAE selected variant `g111`, with a mean of 0.371 eV (95% bootstrap interval [0.357, 0.385] eV) across the five paired repeats. Its locked test MAE was 0.382 eV [0.377, 0.387] eV. The test values were exposed only after the selection rule had fixed the promoted architecture.

The orthogonal contrasts separate evidence for the three prespecified modules from variation between random partitions. Because P jointly changes normalization placement and distance gating, its contrast supports or rejects that composite block only; it does not identify either subchange as the causal source of an effect. On validation MAE, the enabled-minus-disabled main-effect contrasts were G: $\Delta = -0.018$ [-0.020, -0.016] eV (supported reduction); E: $\Delta = -0.003$ [-0.010, +0.004] eV (inconclusive); P: $\Delta = -0.022$ [-0.029, -0.018] eV (supported reduction). Directional validation intervals were also observed for $G : P$: $\Delta = -0.007$ [-0.011, -0.003] eV (supported reduction); $E : P$: $\Delta = +0.005$ [+0.000, +0.010] eV (supported increase). All other prespecified interaction intervals included zero. The repeat-level contrast agreed with the pooled point-estimate direction in G 5/5, P 5/5, $G : P$ 5/5, $E : P$ 4/5 paired repeats. With five paired repeats, the percentile intervals and sign counts are descriptive uncertainty summaries rather than large-sample hypothesis tests.

B. Interpolation, chemical transfer, and low-energy behavior

Using the promoted architecture, pooled DART test MAEs were 0.418 eV for random out-of-fold prediction, 0.450 eV when host–impurity pairs were held out, 0.938 eV for held-out hosts, 0.844 eV for held-out impurities, and 0.386 eV for the predefined chemistry block. Table II compares the aligned DART, periodic SchNet, and validation-selected descriptor predictions. The paired intervals, rather than unpaired differences between aggregate scores, determine whether a directional model difference is supported. For Random OOF, the paired comparator-minus-DART absolute-error differences were SchNet $\Delta = +0.154$ [+0.136, +0.174] eV (supports lower DART error) and LightGBM $\Delta = +0.208$ [+0.194, +0.221] eV (supports lower DART error). For Pair OOF, the paired comparator-minus-DART absolute-error differences were SchNet $\Delta = +0.178$ [+0.157, +0.202] eV (supports lower DART error) and LightGBM $\Delta = +0.224$ [+0.207, +0.242] eV (supports lower DART error). For Host OOF, the paired comparator-minus-DART absolute-error differences were SchNet $\Delta = +5.765$ [+3.006, +9.296] eV (supports lower DART error) and Histogram GB $\Delta = +0.335$ [+0.210, +0.461] eV (supports lower DART error). For Impurity OOF, the paired comparator-minus-DART absolute-error differences were SchNet $\Delta = +0.848$ [+0.626, +1.092] eV (supports lower DART error) and Histogram GB $\Delta = +0.487$ [+0.363, +0.619] eV (supports lower DART error). For Chemistry block, the paired comparator-minus-DART absolute-error differences were SchNet $\Delta = +0.122$ [+0.069, +0.175] eV (supports lower DART error) and LightGBM $\Delta = +0.256$ [+0.189, +0.323] eV (supports lower DART error).

Aggregate MAE can hide failures in chemically sparse or screening-relevant subsets. Table III therefore reports equal-weight host and impurity macro errors together with performance on nonpositive formation energies and the lowest-energy decile. For example, on pair-held-out predictions, the true low-decile MAE was 0.480 eV and its recall by the predicted low decile was 86.0%. None of these low-energy summaries entered architecture or hyperparameter selection.

C. Descriptive error heterogeneity

Across host and impurity groups, the Spearman associations between group sample count and group MAE were -0.523 and -0.001, respectively. The largest group MAEs occurred for host Bi2I6 (1.851 eV; $n = 52$) and impurity In (1.502 eV; $n = 35$). Because group size covaries with chemistry, these are descriptive associations rather than evidence that sample count alone causes the observed errors.

D. Held-out uncertainty and selective prediction

On the untouched uncertainty test partition, the five-member ensemble had MAE 0.377 eV, Gaussian negative log likelihood 1.128, and mean Gaussian CRPS 0.324 eV. The Spearman correlation between calibrated standard deviation and absolute error was 0.546. The risk–coverage curve had AURC 0.177 eV and excess AURC 0.095 eV relative to oracle error ordering.

At 90% nominal coverage, the split-conformal interval covered 87.2% of test structures with mean width 1.141 eV. Table IV reports all prespecified nominal levels. These are marginal held-out coverage results; they do not establish conditional coverage for every host or impurity.

E. Out-of-fold screening decisions within IMP2D

The host–impurity-pair folds provide one out-of-fold prediction for every retained structure. Across pairs containing both incorporation classes, DART recovered the lower-energy class with 91.0% accuracy; the predicted and reference class margins had Spearman correlation

TABLE I. Paired 2^3 architecture factorial. Values are mean MAE with 95% bootstrap intervals over five paired repeats, in eV. G denotes gated readout, E defect-environment enrichment, and P the composite pre-norm distance-gated local block. Bold marks the validation-selected variant; test MAE was not used in its selection.

Variant	G	E	P	Validation MAE	Locked test MAE
g000	0	0	0	0.422 [0.413, 0.430]	0.419 [0.410, 0.433]
g001	0	0	1	0.394 [0.375, 0.413]	0.402 [0.393, 0.413]
g010	0	1	0	0.404 [0.390, 0.417]	0.412 [0.405, 0.421]
g011	0	1	1	0.401 [0.396, 0.407]	0.410 [0.394, 0.429]
g100	1	0	0	0.401 [0.389, 0.413]	0.413 [0.407, 0.420]
g101	1	0	1	0.374 [0.357, 0.388]	0.387 [0.375, 0.400]
g110	1	1	0	0.404 [0.396, 0.410]	0.405 [0.394, 0.415]
g111	1	1	1	**0.371 [0.357, 0.385]**	**0.382 [0.377, 0.387]**

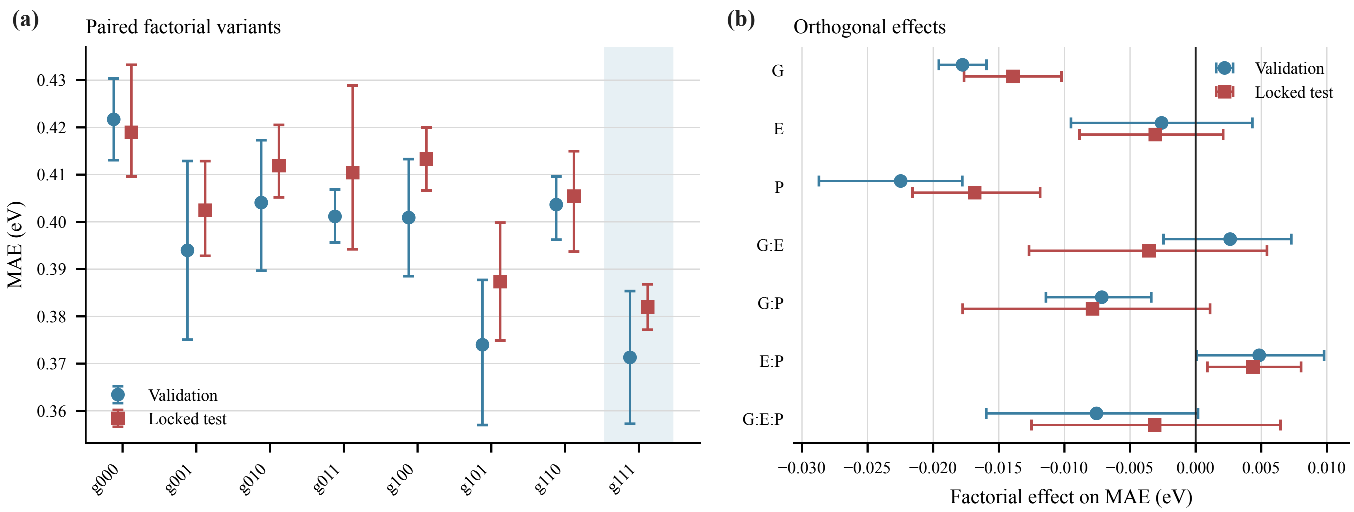

FIG. 2. Paired architecture experiment. (a) Validation and locked-test MAE for all eight G/E/P variants; shading marks the validation-selected variant. (b) Orthogonal main effects and interactions on validation and locked-test MAE, with 95% bootstrap intervals over paired repeats. Negative effects reduce MAE.

0.954 and margin MAE 0.749 eV. Selecting one configuration globally gave 69.5% exact-site accuracy and mean regret 0.139 eV. When the incorporation class was fixed, exact and top-two site accuracies were 73.6% and 91.4%, respectively, with mean regret 0.103 eV.

V. DISCUSSION

A. What the paired factorial establishes

The factorial addresses a narrower and more defensible question than a collection of sequential ablations: whether each prespecified module changes error after averaging over the other two modules on the same data partitions. The validation-selected configuration, `g111`, is therefore tied to a reproducible selection rule rather than to its best observed test split. Locked-test contrasts are useful confirmation of direction and magnitude, but they do not retroactively participate in architecture choice. Likewise, the P factor remains a composite intervention, so any mechanistic attribution to pre-normalization or distance gating separately would require another factorial. The validation contrasts support error reductions from G and P, but not an independent main effect of E; moreover, the G:P interaction reduces MAE whereas the E:P interaction increases it. Selection of the all-enabled variant should therefore be read as the outcome of the prespecified predictive ordering rule, not as evidence that every enabled module is independently beneficial.

B. Transfer results define a bounded applicability domain

Random out-of-fold prediction (0.418 eV) and the grouped protocols answer different questions. Pair-, host-, impurity-, and chemistry-block MAEs were 0.450, 0.938, 0.844, and 0.386 eV, respectively. These values should not be collapsed into one headline score: holding out an entire host probes a different extrapolation axis from holding out an impurity or only their combination. The paired confidence intervals against SchNet and the se-

TABLE II. Pooled out-of-fold or single-block test MAE (eV). The descriptor column uses the family selected separately within each regime by validation MAE (Histogram GB, LightGBM in the resulting selections). Δ is comparator minus DART absolute error with a paired 95% bootstrap interval using the regime-matched resampling unit; positive values favor DART.

Regime	DART	SchNet	Descriptor	Δ_{SchNet}	$\Delta_{\mathrm{desc.}}$
Random OOF	0.418	0.571	0.625	+0.154 [+0.136, +0.174]	+0.208 [+0.194, +0.221]
Pair OOF	0.450	0.628	0.674	+0.178 [+0.157, +0.202]	+0.224 [+0.207, +0.242]
Host OOF	0.938	6.703	1.273	+5.765 [+3.006, +9.296]	+0.335 [+0.210, +0.461]
Impurity OOF	0.844	1.692	1.331	+0.848 [+0.626, +1.092]	+0.487 [+0.363, +0.619]
Chemistry block	0.386	0.508	0.642	+0.122 [+0.069, +0.175]	+0.256 [+0.189, +0.323]

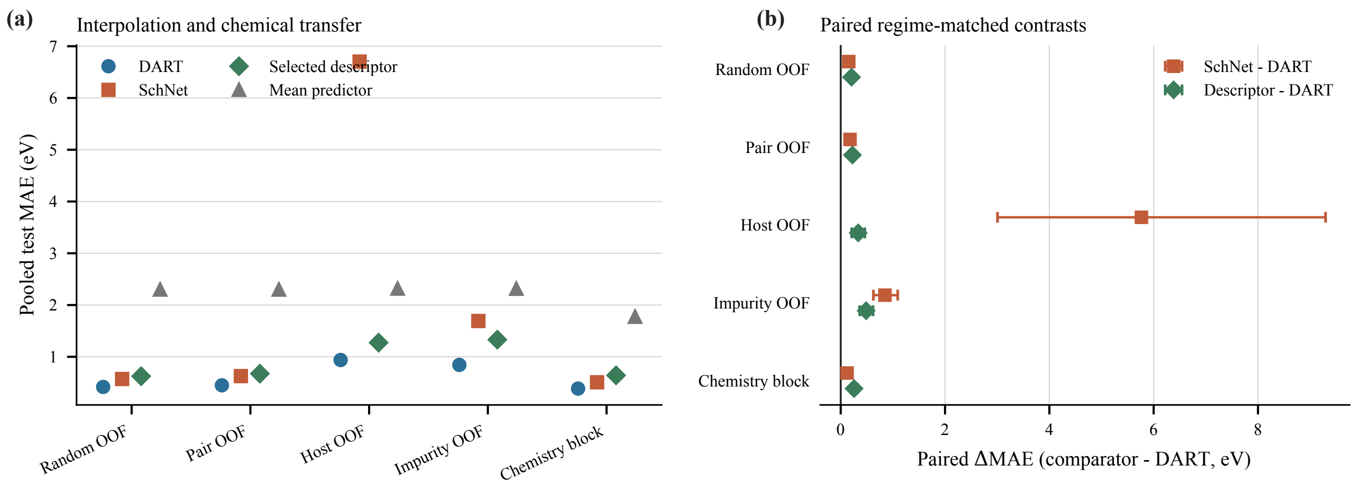

FIG. 3. Interpolation and chemical-transfer evaluation. (a) Pooled test MAE for DART, periodic SchNet, the descriptor family selected within each regime using validation data, and a training-target mean predictor. (b) Paired differences in absolute error relative to DART with 95% bootstrap intervals using the regime-matched resampling units defined in Sec. III A. Positive values favor DART.

lected descriptor baseline further distinguish a supported model difference from a small aggregate fluctuation.

The DART–SchNet contrast is deliberately end to end. DART includes fixed ct-UAE vectors and the JARVIS-initialized input and local blocks, whereas SchNet uses a learned atomic-number embedding without that initialization. The paired differences therefore compare the two prespecified learning recipes under aligned samples and splits; they cannot assign any difference to the backbone architecture, input representation, initialization, or graph readout alone.

All grouped tests remain internal to the 44-host, 65-impurity IMP2D chemical universe. They therefore quantify transfer across defined partitions of this database, not generalization to arbitrary 2D compounds, charge states, defect families, or chemical-potential conventions. Moreover, the JARVIS source task contains reduced-formula matches for 26 IMP2D host labels and contains every IMP2D impurity element. The grouped results therefore test transfer without IMP2D target labels for the held-out chemistry, not de novo extrapolation from an element- or formula-disjoint pretraining corpus. Macro errors and low-energy diagnostics are important because a favorable pooled MAE can coexist with poor performance for rare hosts or the candidates most relevant to screening.

C. Uncertainty supports abstention, not universal reliability

At 90% nominal coverage, observed marginal coverage was 87.2% with mean interval width 1.141 eV. Coverage and width must be interpreted together: a wide interval can satisfy coverage while offering little screening resolution. Error discrimination is separately summarized by the uncertainty–error correlation (0.546) and excess AURC (0.095 eV). The oracle gap in Fig. 5 makes explicit how much rejection performance remains unrecovered.

The conformal guarantee depends on exchangeability between the calibration and test examples. It is marginal rather than host-conditional or impurity-conditional, and it does not automatically extend to the grouped transfer regimes. The ensemble spread should consequently be treated as a calibrated model-disagreement score for selective prediction, not as a complete physical uncertainty decomposition.

TABLE III. Applicability-domain diagnostics for DART. Macro MAEs weight each host or impurity equally. Nonpositive-target MAE uses $E_f \leq 0$; low-decile MAE and recall use the true and predicted lowest 10% of each pooled test regime. Energy errors are in eV and recall is in percent. A dash denotes a regime with no eligible nonpositive test targets.

Regime	Host macro	Impurity macro	Nonpositive-target MAE	Low-decile MAE	Low-decile recall
Random OOF	0.489	0.440	0.396	0.440	87.1
Pair OOF	0.533	0.468	0.450	0.480	86.0
Host OOF	0.990	0.968	1.134	1.284	69.2
Impurity OOF	0.887	0.826	0.973	1.041	71.6
Chemistry block	0.384	0.387	–	0.305	75.0

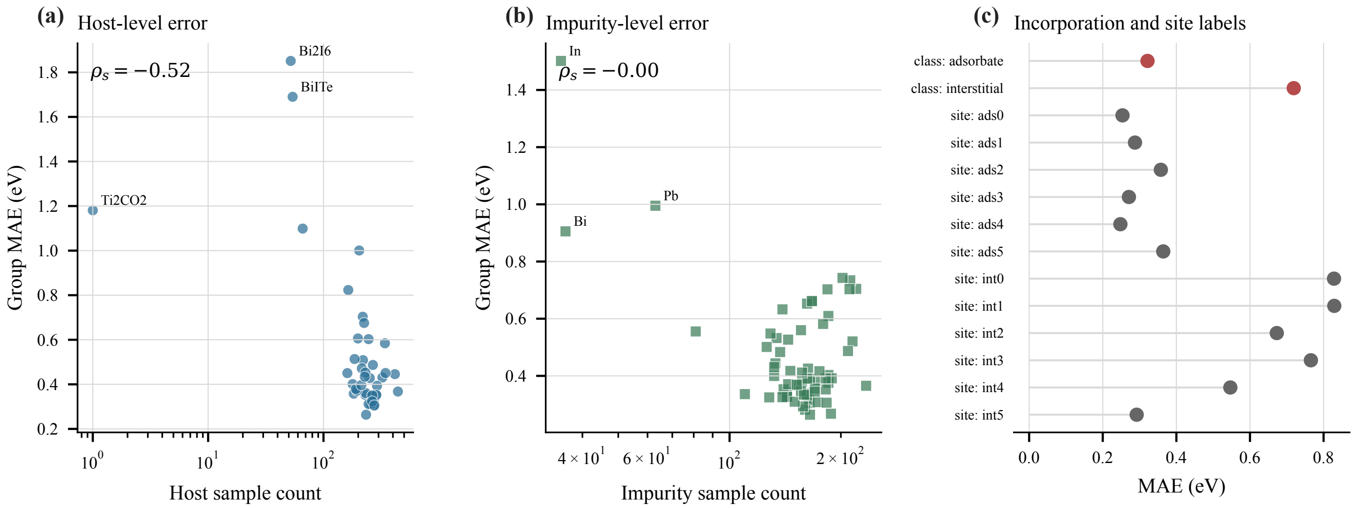

FIG. 4. Error heterogeneity from DART host–impurity-pair-held-out out-of-fold predictions. (a,b) Host- and impurity-group MAE versus the number of retained structures; labels mark the three largest group MAEs and ρ_s is the descriptive Spearman association. (c) MAE by incorporation class and source site label. These summaries were not used for architecture, hyperparameter, or model selection.

TABLE IV. Held-out split-conformal interval performance on the frozen test structures. Coverage is in percent and mean width is in eV.

Nominal	Observed	Mean width	Quantile
50	52.0	0.336	0.270
80	78.3	0.752	0.604
90	87.2	1.141	0.917
95	93.7	1.847	1.484

D. Screening utility and interpretation boundary

The out-of-fold materials analysis converts per-structure regression into decisions that a screening workflow actually makes. Incorporation-class accuracy (91.0%) tests whether adsorbate and interstitial minima are ordered correctly, while exact-site accuracy and regret distinguish recovering the precise minimum from choosing a near-degenerate alternative. The within-class top-two result (91.4%) additionally measures whether a small follow-up queue retains the reference minimum.

These are retrospective decisions against IMP2D labels. They do not establish thermodynamic prevalence, kinetic accessibility, synthesis feasibility, or a new materials discovery. No independently generated first-principles test set matched to the canonical target definition is evaluated.

E. Limitations and next experiments

The evidence is limited to neutral adsorbates and interstitial impurities and to the source database's reference energies and structural relaxations. Vacancies, substitutions, charged defects, transition levels, migration barriers, finite-temperature free energies, and competing phases are outside scope. Duplicate control removes numerical equivalents detectable by the two declared fingerprints but cannot prove that every physically redundant configuration has been identified. The impurity-identity audit also removes 349 rows in which the impurity element is a constituent of the host because the relaxed data do not preserve which same-element atom was inserted. The present conclusions therefore exclude those compositionally ambiguous cases; retaining them would require a persistent atom mapping from structure generation through relaxation. SchNet and the descriptor

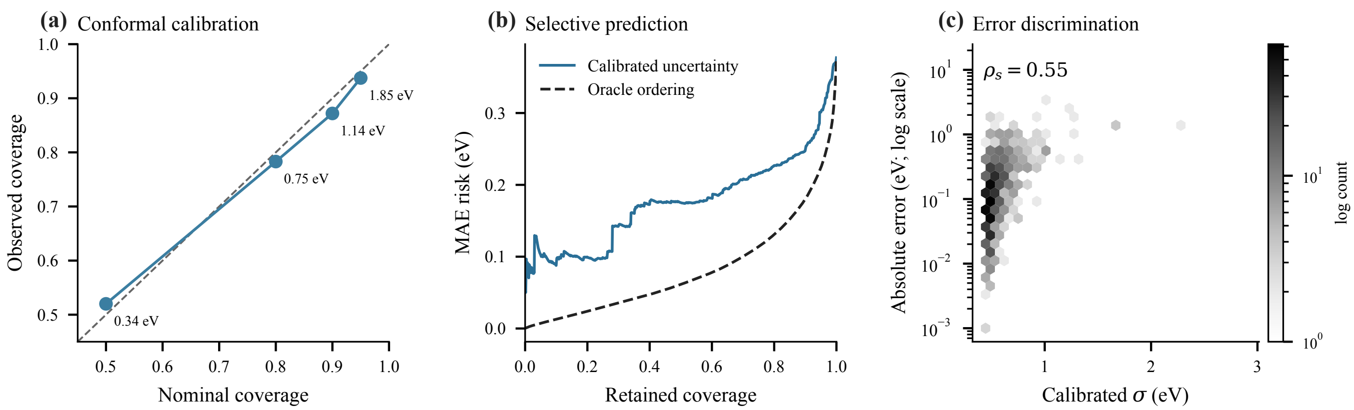

FIG. 5. Held-out uncertainty evaluation. (a) Nominal versus observed split-conformal coverage; annotations give mean interval widths. (b) MAE risk as increasingly uncertain predictions are rejected, compared with oracle ordering by realized error. (c) Calibrated standard deviation versus absolute error on the untouched test partition.

TABLE V. Out-of-fold screening metrics from host–impurity-pair folds. Intervals are 95% nonparametric cluster-bootstrap intervals over host–impurity pairs; multiple within-class decisions from one pair remain together in every resample.

Metric	Mean	95% interval	n
Incorporation-class preference accuracy (%)	91.0	[89.7, 92.4]	1730
Global exact-site accuracy (%)	69.5	[67.3, 71.7]	1730
Global screening regret (eV)	0.139	[0.113, 0.169]	1730
Within-class exact-site accuracy (%)	73.6	[72.0, 75.1]	3140
Within-class top-2 accuracy (%)	91.4	[90.2, 92.6]	2145
Within-class screening regret (eV)	0.103	[0.087, 0.121]	3140

families are strong split-aligned comparators, not an exhaustive benchmark of all atomistic architectures.

The highest-value extension is a preregistered external validation set whose structures, chemical potentials, convergence settings, and target formula are matched exactly to the training evidence. Such calculations are deliberately not claimed here. A second useful extension would calibrate and evaluate uncertainty separately on chemically shifted partitions, where the exchangeability assumption is intentionally stressed.

VI. CONCLUSION

We established a provenance-audited evaluation of a defect-aware graph model for neutral impurity incorporation energies in two-dimensional materials. A paired 2^3 experiment selected `g111` using validation data only, after which the same architecture was evaluated on random, pair-, host-, impurity-, and chemistry-block partitions against split-aligned SchNet and descriptor baselines. The corresponding DART MAEs were 0.418, 0.450, 0.938, 0.844, and 0.386 eV.

On a separate held-out uncertainty split, 90% conformal intervals achieved 87.2% marginal coverage at mean width 1.141 eV. Pair-held-out predictions recovered the preferred incorporation class with 91.0% accuracy and incurred mean global site-selection regret 0.139 eV. Together, these results define both the predictive utility and the observable failure modes of the surrogate within IMP2D. They do not constitute prospective first-principles validation or evidence for materials outside the audited chemical domain.

DATA AND CODE AVAILABILITY

The source IMP2D data are available as version 2 of the Interstitial and Adsorbate Structure Database [25]; their construction is described in Ref. [1]. The exact sample audit, immutable split definitions, resolved experiment configurations, analysis scripts, and machine-readable result bundles for this study are available at https://github.com/chimeraHHH/2d-defect-dast. A release-specific commit and archival identifier will be added before publication.

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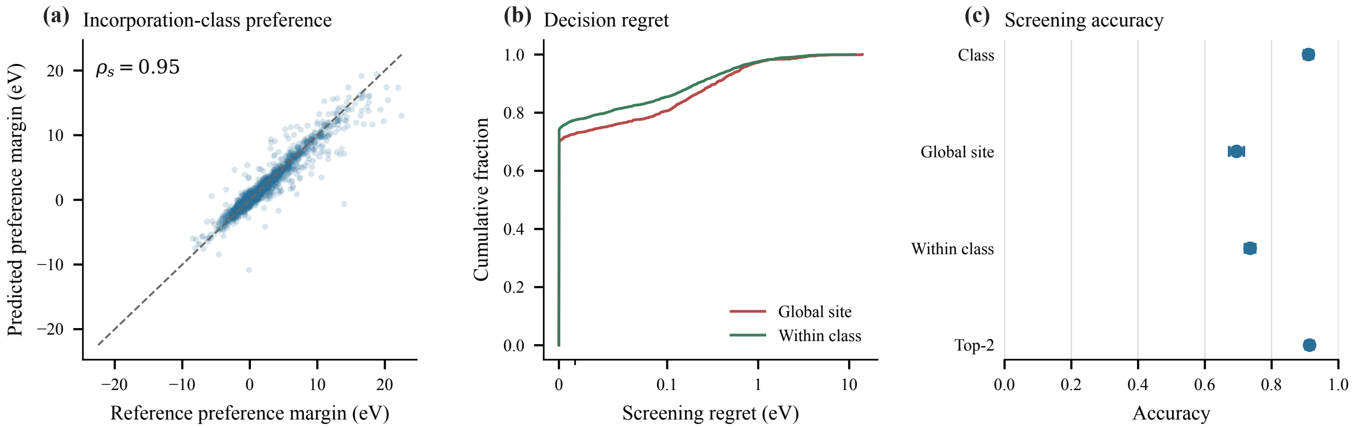

FIG. 6. Screening analysis from pair-held-out predictions. (a) Predicted versus reference preference margins between adsorbate and interstitial configurations. (b) Empirical cumulative distributions of global and within-class screening regret. (c) Host–impurity-pair cluster-bootstrap intervals for incorporation-class, exact-site, and top-two decision accuracy.

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